DCPL AI-Tester

Technology Stack & Local Infrastructure Deployment Runbook
Prepared for Dimakh Consultants Support Team

1. Application Information

Component	Technology / Framework
Frontend	Next.js (React Framework, v16.2.7)
Backend API	FastAPI (Python, v0.136.3), Uvicorn (v0.48.0)
Audit Engine	Playwright (v1.60.0, Chromium browser), BeautifulSoup4 (v4.14.3)
Repository	https://github.com/Adi070106/DCPL_AI-Tester.git

2. Infrastructure Requirements

- **Host Operating System:** Ubuntu (20.04 LTS / 22.04 LTS / 24.04 LTS) or Windows Server.
- **Runtimes Required:** Docker Daemon and Node.js (v18.x or v20.x) installed on the server.
- **Database:** SQLite (Default). Database runs automatically out of a single file (app.db) inside the backend container. No dedicated database service (MySQL/PostgreSQL) setup is required.

3. Environment Configurations

Before deployment, create the following configuration files on the server.

Backend Configuration File: /backend/.env

```
DATABASE_URL=sqlite:///./app.db
JWT_SECRET=super_secret_key_for_website_qa_tester_2026
CORS_ORIGINS=http://localhost:3000,http://:3000
```

Frontend Configuration File: /frontend/.env.production

```
NEXT_PUBLIC_API_URL=http://:8000
```

Note: Replace <server-ip> with the corporate server's IP address or internal network domain.

4. Build & Deployment Commands

Step 1: Deploy Backend (Dockerized)

Docker handles all Playwright system dependencies automatically.

```
# 1. Navigate to the backend directory
cd /backend

# 2. Build the Docker container image
docker build -t dcpl-tester-backend .

# 3. Run the container with persistent storage volume mapped
docker run -d \
  --name dcpl-backend \
  -p 8000:8000 \
  -v /var/data/dcpl-storage:/app/storage \
  --env-file .env \
  dcpl-tester-backend
```

Step 2: Deploy Frontend (Node.js PM2)

```
# 1. Navigate to the frontend directory
cd /frontend

# 2. Install dependencies & build production assets
npm install
npm run build

# 3. Serve the Next.js server permanently using PM2 on port 3000
npm install -g pm2
pm2 start npm --name "dcpl-frontend" -- run start -- -p 3000
pm2 save
pm2 startup
```